



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

CS2002-1 – OBJECT ORIENTED PROGRAMMING

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The new keyword is used to
 - A) Create a new class
 - B) Allocate memory for an object
 - C) Call a method
 - D) Declare a variable
2. Methods cannot have:
 - A) Parameters
 - B) Return values
 - C) Statements
 - D) Main function
3. Method overloading is
 - A) Having multiple methods with the same name but different parameters
 - B) Having multiple methods with the same name but different return types
 - C) Having multiple methods with the same name and parameters
 - D) Having multiple methods with different names but the same parameters
4. How can you access private members of a class from outside the class?
 - A) Directly
 - B) Using public methods
 - C) Using private methods
 - D) It's not possible
5. Which of the following is the correct way to declare an object of a class named Car?
 - A) Car car = Car();
 - B) Car car = new Car();
 - C) Car car = create Car();
 - D) Car car = new Object();
6. What is the result of modifying an object inside a method when it is passed as an argument?
 - A) A new object is created and modified.
 - B) The method throws an exception.
 - C) The original object is modified.
 - D) The object cannot be modified.
7. Which of the following is a correct way to pass an object as an argument to a method?
 - A) method(MyClass obj);
 - B) method(int obj);
 - C) method("obj");
 - D) method(new MyClass());
8. Which keyword restricts access to a class member to within the same class only?
 - A) public
 - B) protected
 - C) private
 - D) static
9. What does the 'final' keyword do when applied to a class?
 - A) Prevents the class from being instantiated.
 - B) Prevents the class from being subclassed.
 - C) Prevents the class members from being modified.
 - D) Indicates that the class is abstract.
10. Which of the following statements about 'static' variables is true?
 - A) They are specific to each instance of a class.
 - B) They are shared across all instances of a class.
 - C) They must be initialized within the constructor.
 - D) They can be declared with the 'final' keyword only.
11. Which of these can be used to fully abstract a class from its implementation?
 - A) Objects
 - B) Packages
 - C) Interfaces
 - D) class



- ### PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Apply the concept of class fundamentals to design a simple Java class representing a hospital patient, including relevant fields and methods. Assume suitable attributes, develop add and display functions and a constructor.	8	L*3	1	1
	b) Describe how the abstract and final keywords are used in Java.	8	L2	2	1
2.	a) Demonstrate how to use constructors and overloading of constructors by creating a class Rectangle that initializes length and breadth through a constructor with parameters and without. Write a method that calculates the area.	8	L3	2	1
	b) Use the static keyword to create a class-level counter that tracks the number of objects created for a Patient class.	4	L3	1	1
	c) Explain the use of super keyword in Java.	4	L2	2	1
3.	a) Create a program to demonstrate how constructors are called in a multilevel inheritance hierarchy involving classes Person → Employee → Manager. Assume suitable methods and variables to demonstrate the initialization of values.	8	L3	2	1
	b) Apply the concept of method overriding by creating a superclass Shape and a subclass Circle that overrides the area() method.	8	L3	2	1

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | Explain concept of Multiple Catch clauses in Java. Write a Java Program that demonstrates handling of Arithmetic exception and Array Index Out of Bands Exception under single try block. | 8 | L2 | 3 | 1 |
| b) | Implement a Java program that creates three threads by implementing the Runnable interface. | 8 | L3 | 3 | 1 |
| 5. a) | Write a program that uses throw and throws to create an exception and throws it for checking if a person is eligible to vote. | 8 | L3 | 3 | 1 |
| b) | Explain any four file handling functions of Java. | 8 | L2 | 4 | 1 |
| 6. a) | Use the concept of serial access files to develop a Java program that creates a text file students.txt, writes student names into it, and then reads them sequentially. | 8 | L3 | 4 | 1 |
| b) | Explain the syntax of exception handling in Java. Demonstrate with an illustration. | 8 | L2 | 3 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | Explain the syntax and usage of any four swing objects that are used for creating graphical user interface in Java. | 8 | L2 | 5 | 1 |
| b) | Apply the List interface by writing a program that stores student names in an ArrayList and displays them in both insertion order and sorted order. | 8 | L3 | 5 | 1 |
| 8. a) | Develop a Swing program by creating a simple GUI application that takes user input through a JTextField and displays it in a JLabel when a button is clicked. | 8 | L3 | 5 | 1 |
| b) | Demonstrate the usage of different methods on a Linked List that holds string objects. | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
